

MNHack8 (8th MareNostrum Hackathon)

SEU Fault-Tolerance Characterization Across Classical and Quantum GPU Substrates

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Abstract

Hardware ECC on modern GPUs protects DRAM but leaves registers, cache, shared memory, and in-transit data uncovered, creating a real and largely uncharacterized risk of silent data corruption from radiation-induced Single Event Upsets (SEUs). This project characterizes how such faults propagate differently through a classical GPU-accelerated finite element solver versus a GPU-simulated quantum statevector evolution of the same underlying PDE, using a scale-matched comparison to avoid the invalid move of comparing systems of mismatched size. A separate classical-only scaling study on MareNostrum 5 establishes how propagation behavior changes with problem scale, and a hybrid-pipeline study characterizes which classical-quantum data handoff points are least protected by existing fault-tolerance mechanisms – a gap explicitly left open by existing formal fault-tolerance frameworks for hybrid scientific workflows.

Chapter 1

Introduction and Motivation

1.1 Project Overview

This project characterizes how radiation-induced bit-flip faults (Single Event Upsets, SEUs) propagate through a classical GPU-accelerated finite element solver and a GPU-simulated quantum statevector evolution of the same underlying PDE. It extends the assembly-free FEM heat solver built and optimized at MNHack7, which achieved a reduction from roughly 5 h (naive, CPU-only) to roughly 1.78 s (optimized, CPU+GPU) on MareNostrum 5.

1.2 The Silent Data Corruption Problem

Hardware ECC on GPUs protects DRAM. It does not protect registers, cache, shared memory, or in-transit data – an SEU striking any of these produces a silently wrong numerical result with no crash and no error flag. This gap is the specific motivation for injecting faults below the DRAM level rather than at it: fault injection restricted to global memory alone would strike precisely the level ECC already protects, which is the wrong place to demonstrate the problem this project is actually about.

1.3 Research Question

[TODO: State the one-sentence project description here once finalized. Working version from prior sync: “SEU fault tolerance characterisation for assembly-free FEM solvers and quantum surrogates on MareNostrum 5.”]

1.4 Scope and Limitations

[TODO: Pull this section together once Layers 0–3 have real results – premature to write a scope/limitations statement before the verification layer (chapter 2) has confirmed the baseline solver is even correct.]

1.5 Related Work

[TODO: Every entry below needs verification against a live source before this chapter is considered submission-ready. Confidence levels are an honest assessment from the

last working session, not a substitute for checking – do not cite any of these in an actual submission without confirming them first.]

- Algorithm-based fault tolerance via checksums: the foundational reference. High confidence it exists roughly as cited [?].
- Physics of radiation-induced soft errors in semiconductor devices: standard reference. High confidence [?].
- Field-defining survey on silent data corruption at exascale. High confidence [?].
- Self-stabilizing / fault-tolerant iterative solvers – the closest existing work to this project’s classical control condition. Moderate confidence on exact authorship and title; finding and confirming the correct paper in this line is the single highest-priority literature task before finalizing this chapter [?].
- GPU fault-injection tooling (NVBitFI, built on NVBit; SASSIFI) – worth citing even though hand-rolled injection is used here, for direct control over fault semantics. Moderate confidence [??].
- StreamFlow: explicitly excludes silent errors from its formal fault-tolerance model for hybrid workflows – the specific, citable gap motivating the hybrid-pipeline chapter ([chapter 10](#)).
- Standard grounding for the Kraus-operator/noise-channel treatment needed to state precisely what a quantum-side fault is (??) [?].

Chapter 2

Layer 0 — Verified Reference Solver

Chapter 3

Layer 1 — Heterogeneous Material Fields

Chapter 4

Layer 2 — Fault Model

Chapter 5

Layer 3 — Measurement Protocol

Chapter 6

Layer 4 — Detection Layer

Chapter 7

Layer 5 — Ensemble Driver and Statistics

Chapter 8

Layer 6 — Quantum Substrate

Chapter 9

Layer 7 — Scale-Matched Comparison

Chapter 10

Layer 8 — Hybrid Pipeline and Handoffs

Chapter 11

Layer 9 — MareNostrum 5 Scaling

Chapter 12

Layer 10 — Synthesis and Discussion